PROBLEM Find the value of $(52+6\sqrt{43})^{3/2}-(52-6\sqrt{43})^{3/2}$. **REFERENCE ANSWER** 828

STUDENT Original solution

[37] So $A=(3+\sqrt{43})^2$, and $B=(3-\sqrt{43})^2$

[46] But here, $x=3-\sqrt{43}$, which is negative

[48] $[(3-\sqrt{43})^2]^{3/2}=(3-\sqrt{43})^3$

✂ ---------- *cut point*

[68] $(3+\sqrt{43})^3-(3-\sqrt{43})^3 = \cdots = 140\sqrt{43}$

[69] Answer: $140\sqrt{43}$ ✗

→

TEACHER Critique

[48] $[(3-\sqrt{43})^2]^{3/2}=(3-\sqrt{43})^3$

Fix: This step is mathematically invalid. The expression $(x^2)^{3/2}$ equals $|x|^3$, not x^3, because the square root function returns the non-negative root. [...] The correct simplification is $\left((3-\sqrt{43})^2\right)^{3/2}=(\sqrt{43}-3)^3$, not $(3-\sqrt{43})^3$. [...]

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STUDENT Rewrite

from [48] with the critique

[...] $(x^2)^{3/2}=|x|^3$, so we take absolute value of $(3-\sqrt{43})$ and cube it. Since $3-\sqrt{43}<0$, $|3-\sqrt{43}|=\sqrt{43}-3$, so $\left((3-\sqrt{43})^2\right)^{3/2}=(\sqrt{43}-3)^3$

$A^{3/2}-B^{3/2}=(3+\sqrt{43})^3-(\sqrt{43}-3)^3$

[...] $(3+\sqrt{43})^3=414+70\sqrt{43}$,

$(\sqrt{43}-3)^3=70\sqrt{43}-414$

$(414+70\sqrt{43})-(70\sqrt{43}-414)=828$

Answer: 828 ✓